

1. Administrative & Study Identification

Full Study Title: A Randomized, Double-blind, Parallel Group, Placebo-controlled, Multicenter Phase 3 Trial to Evaluate Efficacy, Safety and Tolerability of Ianalumab on Top of Standard-of-care Therapy in Participants With Active Lupus Nephritis (SIRIUS-LN) **Short Title/Acronym:** SIRIUS-LN **Protocol Number:** CVAY736K12301 **NCT Number:** NCT05126277 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Ianalumab (VAY736) **Phase of Development:** Phase III **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the clinical efficacy of ianalumab compared to placebo (both on top of standard-of-care therapy) in achieving stable Complete Renal Response (CRR) at 60 weeks in participants with active lupus nephritis (LN). **Secondary Objectives:** To evaluate the safety and tolerability of ianalumab, to compare the efficacy of two different dosing regimens (monthly vs. quarterly), and to assess sustained steroid reduction and long-term renal outcomes.

2.2 Background & Rationale To address the unmet need for achieving complete and sustained renal response in adult patients with active lupus nephritis. Current standard-of-care therapies (SoC) have variable efficacy and significant side effects. Ianalumab is a novel B-cell depleting monoclonal antibody targeting the BAFF receptor, potentially offering a more effective long-term disease control by targeting the underlying B-cell-driven autoimmune response in LN.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Parallel Group **Blinding:** Double-blind **Randomization:** 1:1:1 (Ianalumab q4w, Ianalumab q12w, or Placebo)

3.2 Planned Enrollment & Duration Target \$N\$: Approximately 462 evaluable participants **Number of Sites:** Multicenter study across approximately 38 countries **Estimated Study Start:** 14-Jul-2022 **Estimated Primary Completion:** 22-Jun-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult male and female participants aged 18 years or older weighing at least 35 kg. 2. Confirmed clinical diagnosis of Systemic Lupus Erythematosus (SLE) according to EULAR/ACR criteria, with a positive ANA titer $\geq 1:80$. 3. Active LN at screening, proven by a renal biopsy within 6 months indicating ISN/RPS class III or IV active glomerulonephritis with or without co-existing class V features, or pure class V membranous LN.

4.2 Exclusion Criteria 1. Chronic infection with hepatitis B (HBV) or hepatitis C (HCV), or active tuberculosis (TB). 2. Prior treatment with belimumab, telitacicept, abatacept, cyclophosphamide, calcineurin inhibitors, or JAK inhibitors within 12 weeks prior to randomization. 3. History of major organ transplant or hematopoietic stem cell/bone marrow transplant.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Participants receive either Ianalumab 300 mg every 4 weeks (q4w), Ianalumab 300 mg every 12 weeks (q12w; with placebo on weeks 4 and 8 to maintain blinding), or Placebo every 4 weeks. **Route of Administration:** Subcutaneous (s.c.) injection. **Duration of Treatment:** 60 weeks for the primary treatment period.

5.2 Concomitant & Prohibited Therapy Permitted: Standard-of-care (SoC) treatments for LN, which include corticosteroids and mycophenolic acid (MPA). **Prohibited:** Other B-cell targeted therapies, biologics (e.g., belimumab), calcineurin inhibitors, or specific immunoglobulins not explicitly permitted as standard of care during the trial period.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -42 to 0	Informed Consent, Clinical/Medical History, Renal Biopsy (if needed), Lab Tests (ANA, eGFR)
Baseline	Day 0	Randomization, First Subcutaneous Dose, Safety Run-in
Interim	Every 4 Weeks	Subcutaneous injection (Ianalumab or Placebo), Safety Labs, Renal Function Assessment, UPCR, Disease Activity Questionnaires
End of Study	Up to Week 170	Final Efficacy Assessment, Safety Follow-Up monitoring for up to 104 weeks post-treatment

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of stable Complete Renal Response (CRR) at Week 60 (or Week 72 per EMA guidelines)

without treatment discontinuation or renal flares. **Measurement Method:** Central laboratory analysis to evaluate an estimated glomerular filtration rate (eGFR) and 24-hour Urine Protein-Creatinine Ratio (UPCR).

7.2 Secondary & Exploratory Endpoints * Safety and tolerability profile of ianalumab in LN patients. * Proportion of patients achieving a sustained steroid reduction. * Total reduction in circulating B-cells via antibody-dependent cellular toxicity.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Method of collection includes frequent outpatient clinic visits, physical examinations, and patient reporting. **Serious Adverse Events (SAEs):** Assessed and documented continuously, with particular monitoring for infections given the B-cell depleting mechanism of ianalumab. **Data Monitoring Committee (DMC):** Independent Data Monitoring Committee (IDMC) overseen by the sponsor for unblinded safety and efficacy tracking.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy measures. Safety Set for evaluating AE/SAE occurrences. **Sample Size Justification:** $N \approx 462$ adequately powers the trial ($\alpha = 0.05$, $Power = 80\%$) to test for superiority of either ianalumab arm versus placebo in achieving stable CRR. **Handling of Missing Data:** Missing data will be handled utilizing Multiple Imputation methods and Non-Responder Imputation (NRI) rules for the primary composite endpoint.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Centralized and regular onsite monitoring visits at study sites. **Audit Trail:** Robust Electronic Data Capture (EDC) system with complete eCRF locking mechanisms and source data verification for the 24-hour UPCR and biopsy readings.

11. Study Identifiers & Governance

Primary ID: CVAY736K12301 **Registry Number:** NCT05126277 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** SIRIUS-LN **Official Regulatory Title:** A Randomized, Double-blind, Parallel Group, Placebo-controlled, Multicenter Phase 3 Trial to Evaluate Efficacy, Safety and Tolerability of Ianalumab on Top of

12. Clinical Framework (Lupus Nephritis Specific)

Primary Condition: Active Lupus Nephritis (LN) **Diagnosis Threshold:** Biopsy-proven ISN/RPS class III, IV, or V active glomerulonephritis and ANA titer $\geq 1:80$ **Medical Methodology:** Standard-of-care (corticosteroids + mycophenolic acid) combined with B-cell depleting biological therapy **Optimization Target:** Complete Renal Response (CRR)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Lupus Nephritis **Education Protocol (HCP):** Site initiation visits and regular protocol safety training **Education Protocol (Patient):** Trial regimen counseling and symptom tracking instructions provided onsite **Quality Audit Mechanism:** Routine laboratory tracking of eGFR and urine protein levels

14. Observation & Follow-Up Schedule

Total Duration: Up to 170 weeks **Onsite Visit Cadence:** Every 4 weeks (q4w) during the 60-week primary treatment phase **Remote Monitoring Frequency:** Continuous pharmacovigilance tracking **Checklist Review Interval:** Every 12 weeks for specific efficacy milestones

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				